Sahaj Bhadja

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Summary

Software Engineer with 2+ years delivering **enterprise application features**, **SQL-backed services**, and **test-driven bug fixes** using Java, Python, JavaScript/TypeScript, and Agile development practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Implemented and refined **system components** on **Azure SQL/PostgreSQL** for 500+ users, reducing latency from **1.2s to 350ms** through targeted query, indexing, and schema improvements.
- Built **validation and release processes** for 5K+ monthly jobs, improving defect detection, code-review readiness, and reliability of production workflows used by cross-functional stakeholders.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **Flask + React application features** handling 10K+ daily requests, converting technical requirements into maintainable APIs, frontend components, and clear implementation specifications.
- Reduced turnaround time by **60%** through **CI/CD quality controls** and concurrent processing safeguards, improving implementation consistency and production behavior under peak usage windows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **Python testing pipelines** over 100K+ trajectory logs, enabling reproducible evaluation, basic performance analysis, and faster resolution of technical issues during iterative development cycles.
- Produced technical artifacts and debugging summaries, improving communication quality in design/code reviews and helping collaborators implement fixes with lower regression risk.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Implemented **API and ETL services** using OOP, AWS Glue, and S3, improving response times by **30%** and strengthening data transfer reliability across integrated systems.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito across three OS environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **REST API and WebSocket modules** supporting eight workflows and 2M+ synchronized records, improving stability and maintainability of enterprise application behavior in production.
- Automated testing and deployments with **Docker** and Jenkins, reducing deployment time by **50%** and strengthening operational consistency across iterative software releases.

Skills

Languages: **Java**, **JavaScript/TypeScript**, **Python**, SQL, C++
Development: Agile/Waterfall familiarity, OOP, data modeling basics, system-component implementation
Web & APIs: React.js, Flask, FastAPI, Node.js, HTML, XML, REST APIs, basic .NET awareness
Data & Storage: PostgreSQL, MySQL, MongoDB, SQL query optimization, data storage subsystem understanding
Quality & Tooling: test-driven development mindset, unit/integration testing, Git/GitHub, CI/CD, bug fixing, code reviews

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built a **SQL-centric application component** across 10+ databases with schema-aware query generation and execution validation, improving developer productivity and reliability of enterprise reporting tasks.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Developed **enterprise workflow modules** supporting 50K+ users and 1M+ transactions with secure APIs, scalable architecture, and maintainable feature evolution in production.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Implemented **operational workflow services** for 100+ daily appointments, adding monitoring and validation checks that improved record retrieval by **2x** and reduced support churn.